

address. The slave sends an ACK after receiving this byte.

7. A stop bit is sent to the slave.

8. After receiving this command, slave DS1307 sets its time to 8:32:59.

To read back the data from the RTC using the microcontroller, the interface protocol is:

1. Start bit is sent from microcontroller IC2 (master) to RTC IC3 (slave).

2. 0xD0 is sent to the slave. This is the address of the RTC. The slave sends an ACK after receiving this byte.

3. 0x00 is sent to the slave, setting the address pointer to 00. The slave sends an ACK after receiving this address byte.

4. 0xD1 is sent to the slave. This is the address of the RTC (as mentioned in its datasheet). Also, the LSB of 0xD1 is '1,' which means it is a reading command for the slave. The slave sends an ACK after receiving this byte.

5. The data (like seconds, minutes and hours) is read from slave address location.

6. Finally, a stop bit is sent to the slave.

7. After this command, time, say, 8:33:10, is read and can be displayed on the LCD module.

The procedure is similar for day, month and year too.

To derive the power supply for the circuit, the 230V AC mains is stepped down by transformer X1 to deliver a secondary output of 9V, 500 mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C1 and regulated by IC1. Capacitor C2 bypasses the ripples present in the regulated supply. LED1 acts as the power indicator and R1 limits the current through LED1. Rechargeable battery BATT.2 is used for power backup. Battery is charged through resistor R8 and diode D5. Resistor R8 limits the charging current. Diode D6 protects the battery from discharging when mains is present.

Data pins D0 through D7 of the LCD are connected to port pins P2.0 through P2.7 of microcontroller IC2, respectively. Control pins register-

select (RS), read/write ($R/\bar{W}$) and enable (E) are connected to port pins P3.5, P3.6 and P3.7, respectively. Preset VR1 is connected to pin 3 of the LCD for contrast control. Power-'on' reset is provided by the combination of resistor R2 and capacitor C3. Switch S20 is used for manual reset. An 11.0592MHz crystal along with two 33pF capacitors provides the basic clock frequency to microcontroller IC2.

A 16-key numeric keypad for time/date entry is connected to port P0 of IC2. To economise on the use of I/O pins, we have used only eight pins for scanning and sensing sixteen keys. The keypad is arranged in a 4x4 matrix. There are four scan lines/pins, which are set in output mode, and four sense keys, which are used as input lines. Switches S17 through S19 are connected to port pins P1.3 through P1.5 of the microcontroller for event logging, display on the LCD and sending to PC's HyperTerminal, respectively.

Port pins P1.0 and P1.1 of the microcontroller are connected to SDA and SCL lines of IC3, respectively. The microcontroller can write the data to and read from the RTC through SCL and SDA lines. RS232 interface is used to display the data on the PC's HyperTerminal using USART transmission.

Working of the circuit is simple. When you power-'on' the system, you will see messages 'Welcome' and 'EFY Time Logger' displayed on the first and second lines of the LCD, respectively. After an interval of a second, messages 'Set Clock' and 'Enter Hour' are displayed on the first and second lines of the LCD, respectively. Enter hours using the numeric keypad. Then it will ask you to enter minutes, seconds, date, month and year one by one. The time and date are displayed on the first and second lines of the LCD after all the entries are done. Pressing of switch S17 is an event, which is stored with date and time in the logs. Last ten logs can be seen on the LCD by pressing switch S18. Last 100 entries can be retrieved using HyperTerminal. Connect the serial port and set up the HyperTerminal. Now if you press S19,

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